day02笔记

一、作业

1、从“城市-区域”字段中拆分两级信息

数据列：`locations = ["北京市-海淀区", "上海市-浦东新区", "广州市-天河区"]`

拆分为两个列表：

→ `cities = ["北京市", "上海市", "广州市"]`
→ `districts = ["海淀区", "浦东新区", "天河区"]`

```python
locations = ["北京市-海淀区", "上海市-浦东新区", "广州市-天河区"]
# 市
cities = []
# 区
districts = []
# 遍历操作
for loc in locations:
    # 北京市-海淀区
    if '-' in loc:
        # 拆包操作
        c, d = loc.split('-')
        cities.append(c)
        districts.append(d)

print(cities)
print(districts)

print('-' * 20)

# 列表
# lis = ['张三', '李四', '坤坤']
# print(lis[0], lis[1], lis[2])

# name1, name2, name3 = ['张三', '李四', '坤坤']
# print(name1, name2, name3)

# 元祖
# t1, t2 = ('张三', '李四')
# print(t1, t2)

# 字符串
# n1, n2, n3 = '123'
# print(n1, n2, n3)
```

2、统计字符串的个数

```python
text = 'abcdffqdafg'
"""
```

```
counter = {
    a: 2,
    b: 1
}
"""
counter = {}
# 遍历文本数据
for char in text:
    # 判断下某个字符是否在字典中存在，把字符串中的字符当成字典中的键来使用 key: value 形式（键值对）
    # char ---> 'a'
    # char ---> 'b'
    # ...
    if char in counter:
        # 说明已经存在
        # counter[char] ---> 值是数字
        counter[char] += 1
    else:
        # 说明不存在，把这个字符串当成字典的键添加进字典，并赋值为1
        counter[char] = 1

print(counter)

# print(counter['a'])
# print(counter['b'])
# print(counter.get('a'))
```

二、字符串练习题

1、从“姓名(英文)”字段提取英文名

数据：`names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]`

提取英文名：`['Tom', 'Lucy', 'Bob']`

```
1、从“姓名(英文)”字段提取英文名
数据：`names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]`
提取英文名：`['Tom', 'Lucy', 'Bob']`
"""
names = ["张三(Tom)", "李四(Lucy)", "王五(Bob)"]
english_names = []
for name in names:
    if '(' in name and ')' in name:
        # find() 查找字符串某个字符的位置，返回的是索引
        # start 表示的就是提取字符串里面字符的开始位置
        start = name.find('(') + 1
        # end 表示的就是提取字符串里面字符的结束位置。因为切片的时候，是不包含结束位置的
        end = name.find(')')
        en_name = name[start:end]
        english_names.append(en_name)

print(english_names) # ['Tom', 'Lucy', 'Bob']
```

2、清洗含单位的数值字段（"100kg" → 100）

数据：`weights = ["100kg", "200 kg", "50.5kg", "300g", "1.5 t"]`

统一转换为“千克”数值（假设：1t = 1000kg, 1g = 0.001kg）：

结果：`[100.0, 200.0, 50.5, 0.3, 1500.0]`

```python
# 脏数据
weights = ["100kg", "200 kg", "50.5kg", "300g", "1.5 t"]
result = []

for w in weights:
    # 解决空格问题
    w = w.replace(' ', '')
    # 单位的转换
    if w.endswith('kg'):
        num = float(w[:-2])
        result.append(num)
    elif w.endswith('g'):
        num = float(w[:-1]) * 0.001
        result.append(num)
    elif w.endswith('t'):
        num = float(w[:-1]) * 1000
        result.append(num)

print(result)
```

3、生成数据脱敏后的展示字段（姓名、手机、邮箱）

原始数据：

```python
data = [
    {"name": "张三丰", "phone": "13812345678", "email": "zhang@example.com"},
    {"name": "李莫愁", "phone": "13987654321", "email": "li@example.org"}
]
```

脱敏规则：

- 姓名：保留姓，名用 `*` 代替 → `"张**"`
- 手机：`138****5678`
- 邮箱：`z***@example.com`

```python
data = [
    {"name": "张三丰", "phone": "13812345678", "email": "zhang@example.com"},
    {"name": "李莫愁", "phone": "13987654321", "email": "li@example.org"}
]

masked_data = []

# d 表示的是列表中的每一个字典
for d in data:
    # 处理姓名
    name = d['name']
```

```python
    masked_name = name[0] + '*' * (len(name) - 1) if len(name) > 1 else name

    # 处理电话号码
    phone = d['phone']
    masked_phone = phone[:3] + '****' + phone[7:] if len(phone) == 11 else phone

    # 处理邮箱地址
    email = d['email']
    # 邮箱格式的简单验证
    if '@' in email:
        local, domain = email.split('@')
        masked_local = local[0] + '***' if len(local) > 1 else local
        masked_email = masked_local + '@' + domain
    else:
        masked_email = email

    # 添加进列表中
    masked_data.append({
        'name': masked_name,
        'phone': masked_phone,
        'email': masked_email
    })

print(masked_data)
```

三、三目表达式

1、名称：三元运算符、条件表达式

2、是对双分支选择结构的简化操作

3、语法

```python
# x = 10
# y = 20
# if x > y:
#     print(x)
# else:
#     print(y)

# x = 30
# y = 20
# if 前面放的是满足条件的结果，else 后面放的是不满足条件的结果
# result = x if x > y else y
# print(result)

# 根据年龄判断是否成年
age = 20
result = '成年人' if age >= 18 else '未成年人'
print(result)
```

四、数据处理练习题

1、在处理文件路径时，可以使用 `.rfind()` 来查找最后一个斜杠的位置，从而提取文件名。

```
# file_path = "/home/user/documents/report.txt"
# 0 从左边往右边进行查找
# print(file_path.find('/'))
# 20 从右边往左边查找
# print(file_path.rfind('/'))
# 注意点：find() 和 rfind() 如果没有查找找对应的字符的位置会返回-1
# print(file_path.find('-'))
# print(file_path.rfind('-'))

file_path = "/home/user/documents/report.txt"
last_index = file_path.rfind("/")
if last_index != -1:
    file_name = file_path[last_index+1:]
    print(file_name)
else:
    print('没有找到')
```

2、从URL中提取参数 `url = "https://api.example.com/v1/users?name=Alice&age=30&city=Beijing"`

需要提取的数据：`name=Alice&age=30&city=Beijing`

输出结果：`{'name': 'Alice', 'age': '30', 'city': 'Beijing'}`

```
url = "https://api.example.com/v1/users?name=Alice&age=30&city=Beijing"
if '?' in url:
    query_str = url.split('?')[1] # name=Alice&age=30&city=Beijing
    # 把字符串转成字典形式
    params = {}
    # 先进行切割处理
    items = query_str.split('&') # ['name=Alice', 'age=30', 'city=Beijing']
    # 遍历操作
    for it in items:
        # name=Alice
        # age=30
        # city=Beijing
        key, value = it.split('=')

        params[key] = value

    print(params) # {'name': 'Alice', 'age': '30', 'city': 'Beijing'}
```

五、字典和列表组合操作

1、数据

```
school_data = {
    "学校名称": "英才实验中学",
    "年级": "高一年级",
```

```
    "班级列表": [
        {
            "班级ID": "G1-01",
            "班级名称": "高一(1)班",
            "班主任": {
                "教师ID": "T001",
                "姓名": "张伟",
                "性别": "男",
                "联系电话": "13800138001",
                "入职年份": 2015,
                "负责科目": ["语文", "德育"]
            },
            "任课教师": [
                {
                    "教师ID": "T002",
                    "姓名": "李芳",
                    "科目": "数学",
                    "职称": "高级教师"
                },
                {
                    "教师ID": "T003",
                    "姓名": "王强",
                    "科目": "英语",
                    "职称": "一级教师"
                },
                {
                    "教师ID": "T004",
                    "姓名": "赵敏",
                    "科目": "物理",
                    "职称": "二级教师"
                }
            ],
            "班级信息": {
                "成立日期": "2024-09-01",
                "最大容量": 45,
                "当前人数": 42,
                "教室编号": "A201",
                "状态": "正常"
            },
            "学生名单": [
                {
                    "学号": "S2024001",
                    "姓名": "陈小明",
                    "性别": "男",
                    "出生日期": "2008-03-15",
                    "家庭信息": {
                        "父亲姓名": "陈建国",
                        "母亲姓名": "李秀英",
                        "联系电话": "13900139001",
                        "家庭住址": "北京市朝阳区XX路XX号"
                    },
                    "选课情况": ["数学", "物理", "英语", "语文", "体育", "美术"],
                    "学期成绩": {
```

```
                "2024秋季学期": {
                    "语文": 88,
                    "数学": 92,
                    "英语": 85,
                    "物理": 89,
                    "体育": 95,
                    "美术": 90
                },
                "2025春季学期": {
                    "语文": 90,
                    "数学": 94,
                    "英语": 88,
                    "物理": 91,
                    "体育": 96,
                    "美术": 92
                }
            },
            "考勤记录": {
                "2024秋季学期": {
                    "出勤天数": 92,
                    "缺勤天数": 3,
                    "迟到次数": 2
                },
                "2025春季学期": {
                    "出勤天数": 88,
                    "缺勤天数": 1,
                    "迟到次数": 0
                }
            },
            "奖惩记录": [
                {"时间": "2024-11", "类型": "奖励", "内容": "期中考试年级前十", "颁发人":
"张伟"},
                {"时间": "2025-03", "类型": "警告", "内容": "未交作业累计3次", "处理人":
"李芳"}
            ]
        },
        {
            "学号": "S2024002",
            "姓名": "王丽丽",
            "性别": "女",
            "出生日期": "2008-07-22",
            "家庭信息": {
                "父亲姓名": "王建国",
                "母亲姓名": "赵美玲",
                "联系电话": "13900139002",
                "家庭住址": "北京市海淀区XX街XX号"
            },
            "选课情况": ["数学", "物理", "英语", "语文", "音乐", "信息技术"],
            "学期成绩": {
                "2024秋季学期": {
                    "语文": 92,
                    "数学": 88,
                    "英语": 94,
```

```
                        "物理": 85,
                        "音乐": 98,
                        "信息技术": 96
                    },
                    "2025春季学期": {
                        "语文": 93,
                        "数学": 90,
                        "英语": 96,
                        "物理": 87,
                        "音乐": 99,
                        "信息技术": 97
                    }
                },
                "考勤记录": {
                    "2024秋季学期": {
                        "出勤天数": 95,
                        "缺勤天数": 0,
                        "迟到次数": 1
                    },
                    "2025春季学期": {
                        "出勤天数": 89,
                        "缺勤天数": 0,
                        "迟到次数": 0
                    }
                },
                "奖惩记录": [
                    {"时间": "2024-12", "类型": "奖励", "内容": "校园歌手大赛冠军", "颁发人":
"团委"},
                    {"时间": "2025-01", "类型": "奖励", "内容": "编程竞赛一等奖", "颁发人":
"王强"}
                ]
            }

        ]
    },

    {
        "班级ID": "G1-02",
        "班级名称": "高一(2)班",
        "班主任": {
            "教师ID": "T005",
            "姓名": "刘洋",
            "性别": "女",
            "联系电话": "13800138005",
            "入职年份": 2018,
            "负责科目": ["英语", "心理"]
        },
        "任课教师": [
            {"教师ID": "T002", "姓名": "李芳", "科目": "数学", "职称": "高级教师"},
            {"教师ID": "T001", "姓名": "张伟", "科目": "语文", "职称": "高级教师"},
            {"教师ID": "T006", "姓名": "孙浩", "科目": "化学", "职称": "一级教师"}
        ],
        "班级信息": {
```

```
                "成立日期": "2024-09-01",
                "最大容量": 45,
                "当前人数": 44,
                "教室编号": "A202",
                "状态": "正常"
            },
            "学生名单": []
            # 学生数据可按需填充
        }
    ],
    "统计信息": {
        "总班级数": 2,
        "总学生数": 42 + 0,  # G1-01有42人，G1-02暂无学生
        "教师总数": 6,
        "最新更新时间": "2025-04-05"
    }
}
```

2、题目

```
# 1. 打印高一(1)班班主任姓名

# 2. 打印陈小明的2025春季学期数学成绩

# 3. 打印王丽丽的家庭联系电话

# 4. 打印高一(1)班所有任课教师姓名

# 5. 统计陈小明获得的奖励次数

# 6、打印陈小明所有学期各科成绩
```